

Week 3 Report: Robust Federated Learning against Adversarial Attacks

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Abstract

This report summarizes the Week 3 implementation and experimental protocol for the project *Robust Federated Learning against Adversarial Attacks*. The objective is to secure a federated learning pipeline against malicious clients that perform data poisoning through label flipping. The baseline FedAvg system is compared against robust aggregation defenses: Krum, coordinate-wise median, and trimmed mean. The experiments use a lightweight CheXpert setup for binary Cardiomegaly classification and evaluate test accuracy, test loss, accuracy drop, attack success rate, and aggregation computation time.

1 Objective

Federated learning (FL) enables multiple clients to train a shared model while keeping their data local. This is especially relevant in medical imaging, where institutions may not be able to share patient data directly. However, FL is vulnerable to malicious participants. A client can poison its local data or manipulate its model update before sending it to the server.

The Week 3 objective is to implement and evaluate an adversarial FL pipeline with the following components:

- Data poisoning through label flipping on a subset of malicious clients.
- Robust aggregation defenses: Krum, coordinate-wise median, and trimmed mean.
- A fixed-seed experimental protocol with five scenarios.
- Metrics for accuracy, loss, attack impact, and aggregation computation time.
- Learning-curve and timing visualizations.

2 Dataset and Baseline Model

The experiments use the lightweight CheXpert subset from the previous project phase. The task is binary classification for the *Cardiomegaly* label. The preprocessing follows the Week 2 baseline:

- Keep only frontal chest X-ray images.
- Replace uncertain labels, encoded as -1 , with 0 .
- Fill missing target labels with 0 .
- Convert images to grayscale.
- Resize each image to 32×32 .

- Flatten each image into a vector of 1024 pixel features.
- Standardize features using a train-only `StandardScaler`.

The model is a lightweight multilayer perceptron compatible with the Week 2 baseline. It contains one hidden layer with 128 units, ReLU activation, dropout, and a single output logit for binary classification. Training uses binary cross-entropy with logits and the Adam optimizer.

3 Federated Learning Setup

The dataset is split into a training set and a held-out test set. The training set is partitioned across simulated clients. In the default Week 3 configuration, there are 10 clients, of which 2 are malicious in attacked scenarios.

Each communication round follows the standard FL procedure:

1. The server sends the current global model to each client.
2. Each client trains a local model for one local epoch.
3. Honest clients train on clean labels.
4. Malicious clients train on poisoned labels in attacked scenarios.
5. Clients return their local model states to the server.
6. The server aggregates client states using the selected aggregation rule.
7. The new global model is evaluated on the held-out test set.

The default experimental configuration is shown in Table 1.

Table 1: Default Week 3 experiment configuration.

Parameter	Value
Random seed	42
Sample size	10,000
Image size	32×32
Number of clients	10
Malicious clients	2
Communication rounds	20
Local epochs	1
Batch size	128
Learning rate	0.001
Trim ratio	0.2
Attack strength	8.0

4 Adversarial Attack

4.1 Label-Flipping Data Poisoning

The main adversarial attack is label flipping. For the binary target label $y \in \{0, 1\}$, a malicious client trains using the poisoned label:

$$y_{\text{poisoned}} = 1 - y. \tag{1}$$

This attack corrupts the local supervision signal while leaving the input image features unchanged. In the FL setting, the server does not directly observe local labels, so malicious clients can perform this attack without changing the server-side code.

4.2 Poisoned Update Amplification

In practice, plain label flipping with only a small malicious minority can be too weak to visibly separate the learning curves. Therefore, the attacked scenarios use a stronger adversarial FL setting: after training on flipped labels, a malicious client amplifies its local model delta before sending it to the server:

$$w_{\text{malicious}} = w_{\text{global}} + \alpha(w_{\text{poisoned}} - w_{\text{global}}), \quad (2)$$

where α is the attack strength. The default value is $\alpha = 8.0$. Setting `--attack-strength 1.0` recovers the original mild label-flipping attack without amplification.

4.3 Optional Model Noise

The implementation also includes an optional model-poisoning helper. If enabled, Gaussian noise is added to a malicious client’s returned model state:

$$w_{\text{malicious}} = w_{\text{local}} + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2). \quad (3)$$

For the main Week 3 protocol, the default attack is label flipping with poisoned update amplification. Gaussian model noise is disabled unless explicitly enabled through command-line arguments.

5 Aggregation Methods

5.1 FedAvg

Federated Averaging computes a sample-size-weighted average of client model parameters. If client k has n_k samples and returns model state w_k , the server computes:

$$w_{\text{global}} = \sum_{k=1}^K \frac{n_k}{\sum_{j=1}^K n_j} w_k. \quad (4)$$

FedAvg is efficient but vulnerable because poisoned client states directly influence the average.

5.2 Krum

Krum is a Byzantine-robust aggregation method. It flattens each client model state and computes pairwise squared Euclidean distances between client states. For each client i , Krum sums the distances to its $K - f - 2$ nearest neighbors, where K is the number of clients and f is the assumed number of malicious clients:

$$s_i = \sum_{j \in N_i} \|w_i - w_j\|_2^2. \quad (5)$$

The selected global state is the client state with the smallest score:

$$w_{\text{global}} = w_{\arg \min_i s_i}. \quad (6)$$

Krum requires:

$$K \geq 2f + 3. \quad (7)$$

With 10 clients and 2 malicious clients, this condition is satisfied.

5.3 Coordinate-Wise Median

Coordinate-wise median stacks all client parameter vectors and computes the median independently for each parameter coordinate:

$$w_{\text{global}}[d] = \text{median}(w_1[d], w_2[d], \dots, w_K[d]). \quad (8)$$

This reduces the effect of extreme malicious parameter values when malicious clients are a minority.

5.4 Trimmed Mean

Trimmed mean also operates coordinate by coordinate. For each parameter coordinate, values are sorted, the smallest and largest fraction are removed, and the remaining values are averaged. With 10 clients and trim ratio 0.2, the two smallest and two largest values are discarded per coordinate.

If T_d is the set of remaining values after trimming coordinate d , then:

$$w_{\text{global}}[d] = \frac{1}{|T_d|} \sum_{v \in T_d} v. \quad (9)$$

6 Experimental Scenarios

The Week 3 experiment script evaluates five scenarios, shown in Table 2.

Table 2: Week 3 experimental scenarios.

Scenario	Aggregation	Attack
A_FedAvg_No_Attack	FedAvg	None
B_FedAvg_LabelFlip	FedAvg	Label flipping + update amplification
C_Krum_LabelFlip	Krum	Label flipping + update amplification
D_Median_LabelFlip	Coordinate-wise median	Label flipping + update amplification
E_TrimmedMean_LabelFlip	Trimmed mean	Label flipping + update amplification

7 Metrics

The following metrics are recorded after every communication round:

- **Test loss:** binary cross-entropy loss on the held-out test set.
- **Test accuracy:** fraction of correctly classified test examples.
- **Accuracy drop:** difference between clean FedAvg accuracy and the scenario accuracy at the same round.
- **Attack success rate:** positive part of the accuracy drop.
- **Aggregation time:** wall-clock time required by the server aggregation method.

The accuracy drop is computed as:

$$\Delta_{\text{acc}}(r) = \text{Acc}_{\text{clean FedAvg}}(r) - \text{Acc}_{\text{scenario}}(r). \quad (10)$$

The attack success rate is:

$$\text{ASR}(r) = \max(0, \Delta_{\text{acc}}(r)). \quad (11)$$

Accuracy Summary

Table 3 summarizes the test accuracy observed for each experimental scenario. The final accuracy corresponds to the last communication round saved in `experiments/week3_results.csv`.

Table 3: Summary of test accuracy by scenario.

Scenario	Min. Acc.	Max. Acc.	Mean Acc.	Final Acc.
A_FedAvg_No_Attack	0.8770	0.8810	0.8794	0.8805
B_FedAvg_LabelFlip	0.1215	0.5015	0.3440	0.2805
C_Krum_LabelFlip	0.8715	0.8805	0.8766	0.8745
D_Median_LabelFlip	0.8760	0.8805	0.8789	0.8790
E_TrimmedMean_LabelFlip	0.8760	0.8810	0.8790	0.8795

The results show that naive FedAvg is strongly affected by the label-flipping and update-amplification attack. Its final accuracy drops to 0.2805, while the robust aggregation methods remain close to the clean FedAvg baseline. Coordinate-wise median and trimmed mean provide the strongest protection in this run, while Krum also limits the attack damage.

Conclusion

Week 3 completes the main adversarial FL implementation. The project now contains a working attack simulation, robust aggregation defenses, a reproducible experiment script, structured result logging, plotting utilities, and a written report. These components allow direct comparison between clean FedAvg, attacked FedAvg, and robust aggregation methods under label-flipping data poisoning.